Qian Hu

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EMPLOYMENT

University of Glasgow **2024 - Present**
Postdoctoral Researcher

EDUCATION

University of Glasgow **2021 - 2024**
PhD in gravitational-wave astronomy
Thesis: Towards high-precision gravitational wave astronomy: robust and efficient data analysis for ground-based detectors. (Link)
Advisor: Dr. John Veitch

University of Western Australia **2019 - 2021**
Undergraduate Research Intern. Research topic: Gravitational wave source localization.
Advisor: Prof. Linqing Wen

University of Science and Technology of China (USTC) **2017 - 2021**
B.S. (Honor) in astrophysics, GPA: 3.90/4.30.
Thesis: Rapid Sky Localization of Gravitational Waves from Compact Binary Coalescences
Advisors: Prof. Wen Zhao & Prof. Linqing Wen

REASEARCH INTERESTS

- **Gravitational-Wave (GW) Astrophysics**
  - Higher-order characteristics from compact binaries: precession, eccentricity, higher modes etc, and their astrophysical implications.
  - Tests of GR with GWs: methods and robustness.
  - Properties of compact objects: spins, equations of state, etc.
  - Data-driven GW waveform modelling: surrogate models and model uncertainties.
- **GW Data Analysis Techniques**
  - Waveform systematics in parameter estimation and ways of mitigating them.
  - Difficult scenarios in the next-generation detectors: overlapping signals, long signals, and large catalogs.
  - Fast parameter estimation (including fast localization) of CBC sources and interactions with multi-messenger astronomy.
  - Application of machine learning to GW data analysis: detection, parameter estimation, and population and cosmological inference.

SELECTED PUBLICATIONS

Full publication list including collaboration papers can be found at: Inspire-HEP, Google Scholar.

- **Qian Hu**, *Hierarchical Subtraction with Neural Density Estimators as a General Solution to Overlapping Gravitational Wave Signals*. arXiv:2507.05209.
- **Qian Hu**, Jessica Irwin, Qi Sun, Chris Messenger, Lami Suleiman, Ik Siong Heng, John Veitch,

Decoding long-duration GW from BNS with machine learning: Parameter estimation and equations of state. Astrophys.J.Lett. 987 (2025) L17.

- **Qian Hu**, John Veitch, *Costs of Bayesian Parameter Estimation in Third-Generation Gravitational Wave Detectors: an Assessment of Current Acceleration Methods*. Phys.Rev.D 112 (2025) 8, 084039.
- **Qian Hu**, John Veitch, *Rapid pre-merger localization of binary neutron stars in third generation gravitational wave detectors*. Astrophys.J.Lett. 958 (2023) 2, L43.
- **Qian Hu**, John Veitch, *Accumulating errors in tests of general relativity with gravitational waves: overlapping signals and inaccurate waveforms*. Astrophys.J. 945 (2023) 2, 103.
- **Qian Hu**, John Veitch, *Assessing the model waveform accuracy of gravitational waves*. Phys.Rev.D 106 (2022) 4, 044042.
- **Qian Hu**, Cong Zhou, Jhao-Hong Peng, Linqing Wen, Qi Chu, Manoj Kovalam, *Semianalytical Approach for Sky Localization of Gravitational Waves*. Phys.Rev.D 104 (2021) 10, 104008.
- **Qian Hu**, Mingzheng Li, Rui Niu, and Wen Zhao. *Joint Observations of Space-based Gravitational-wave Detectors: Source Localization and Implication for Parity-violating Gravity*. Phys.Rev.D 103 (2021) 6, 064057.

SELECTED TALKS & PRESENTATIONS

Seminars

- *Overlapping gravitational wave signals: Impacts, Solutions, and and an Ambiguous Case in Current Observations*, Max Planck Institute for Gravitational Physics (Albert Einstein Institute), Potsdam, Oct 2025.
- *Machine learning as a cornerstone for future gravitational wave observations*, Tsung-Dao Lee Institute, Shanghai Jiao Tong University, Sept 2025.
  Also at International Centre for Theoretical Physics Asia-Pacific, University of Chinese Academy of Sciences, Beijing in Sept 2025.
- *Decoding binary neutron stars with next-generation gravitational wave detectors: localization, parameter estimation, and equations of state*. Kavli Institute for Astronomy and Astrophysics, Peking University, Dec 2024.
  Also at Tsinghua University in Jan 2025; TianQin Center at Sun Yat-Sen University in Jan 2025; Zhejiang University (Zhoushan) in Dec 2024; and University of Science and Technology of China in Dec 2024.
- *Assessing model waveform accuracy for gravitational waves*, TianQin Center (online), June 2022.

Contributed Talks

- *Hierarchical Subtraction with Neural Density Estimators as a General Solution to Overlapping Gravitational Wave Signals*. ET 4th Annual Meeting, Nov 2025, Opatija, Croatia.
  Also at LVK Collaboration Meeting, Sept 2025, Fort Collins, Colorado, US (online).
- *Solving tough inference problems for the next-generation gravitational-wave detectors with machine learning: long signals and overlapping signals*. GR24–Amaldi16, Glasgow, UK, July 2025.
- *Decoding long-duration GW from BNS with machine learning: Parameter estimation and equations of state*. ET 3rd Annual Meeting, Nov 2024, Warsaw, Poland.
  Also at LVK Collaboration Meeting as a poster in Barcelona, Spain in Sept 2024.
- *Realtime pre-merger localization of BNS in 3G GW detectors*, National Astronomy Meeting 2023 (NAM23), July 2023, Cardiff, UK.
  Also presented a poster at GWNext Beijing in Mar 2024 (Poster Prize, 1st place) and ET 3rd Annual Meeting, Warsaw, Poland, Nov 2024.

- *Systematic error accumulation in testing GR: overlapping signals and waveform systematics.* XIII ET Symposium, Cagliari, Italy, May 2023.
- *On the model waveform accuracy of gravitational waves.* BritGrav 2022 (online), Apr 2022.
- *Quantitative measurement of model differences for CBCs.* LVK Collaboration Meeting (online), Mar 2022.

Posters

- *Costs of Bayesian Parameter Estimation in Third-Generation Gravitational-Wave Detectors.* GR24–Amaldi16, Glasgow, UK, July 2025.
  Also at ET 4th Annual Meeting, Opatija, Croatia, Nov 2025

SOFTWARE DEVELOPMENT

- SealGW: A Python package for fast source localization. Implemented in the SPIIR online detection pipeline.
- WaveCheck: Waveform accuracy check pipeline for online parameter estimation runs, currently running for LVK O4.
- river: A Python package for gravitational wave source parameter estimation with normalizing flows.

TEACHING

Gravitational Wave Detection **2024-2025 @UofGlasgow**
Co-lecturer

Astronomy 1 **2022-2024 @UofGlasgow**
Teaching assistant

Physics 1 **2021-2022 @UofGlasgow**
Teaching assistant

Physics experiment software development **2020 Fall @USTC**
National virtual experiment teaching project

Classical Mechanics & Electrodynamics **2020 Fall @USTC**
Teaching assistant

ACADEMIC SERVICE

- **Peer Review**: Referee for *Physical Review D* (3 manuscripts), *The Astrophysical Journal* (1 manuscript), and IOP Conference Series (1 manuscript).
- **Outreach**: Co-presenter at the Edinburgh Science Festival (Family Fun Day, 2025), Glasgow Science Festival (2024), and WorldCon Glasgow (2024).
- **Conference & Departmental Service**: Abstract reviewer for GR24–Amaldi16.

HONORS & AWARDS

- Poster prize (1st place), GWNext Beijing, 2024.
- Honorary Undergraduate, USTC, 2021
- National Scholarship, Ministry of Education of the PRC, 2020
- National Astronomical Observatory Scholarship, National Astronomical Observatories of China, 2020
- Outstanding Student Scholarship, USTC, 2018 & 2019

PROFESSIONAL SKILLS

- **Programming:** Python, C, Cython, Matlab, Mathematica.
- **High-performance computing:** Linux systems, HTCondor, Slurm.
- **Research-relevant software:** Bilby, PyCBC, LALSuite, DINGO, PyTorch.